

Cisco Packet Tracer Project #01

Basic Office LAN

1. Project Overview

This project simulates a small office LAN using one Cisco 2960 switch and four PCs. All PCs use static IPv4 addressing within the same /24 network.

2. Project Objectives

The objectives are to understand basic LAN connectivity, configure IPv4 addresses, understand Layer 2 switching, inspect ARP, and perform connectivity testing and basic troubleshooting.

3. Network Topology

PC-01 → SW1 Fa0/1

PC-02 → SW1 Fa0/2

PC-03 → SW1 Fa0/3

PC-04 → SW1 Fa0/4

SW1 is a Cisco 2960 switch. Ethernet connections use Copper Straight-Through cables.

4. IP Addressing

Device	IP Address	Mask
PC-01	192.168.1.10	255.255.255.0
PC-02	192.168.1.11	255.255.255.0
PC-03	192.168.1.12	255.255.255.0
PC-04	192.168.1.13	255.255.255.0

5. Default Gateway

The default gateway is left empty because all hosts are in the same subnet. A router is not required for communication between these hosts.

6. Verification Commands

`ipconfig` — checks local IPv4 configuration.

`ping 192.168.1.x` — tests IP connectivity.

`arp -a` — displays learned IP-to-MAC mappings.

7. Technical Explanation

When PC-01 communicates with PC-04 on the same subnet, ARP can resolve PC-04's IPv4 address to its MAC address. The switch then forwards Ethernet frames according to its MAC address table. ICMP Echo Request and Echo Reply are used by ping to verify connectivity.

8. Troubleshooting Exercise

A recommended exercise is to intentionally change PC-04's IP address to another subnet and observe the failed ping. Diagnose the problem using ipconfig, ping, arp -a, cable/link status, and IP/subnet verification.

9. Expected Result

All PCs should be able to communicate with one another because they share the 192.168.1.0/24 network and are connected through SW1.

10. Skills Demonstrated

LAN design, IPv4 addressing, subnetting fundamentals, Ethernet switching, ARP, ICMP, Cisco Packet Tracer, and basic network troubleshooting.